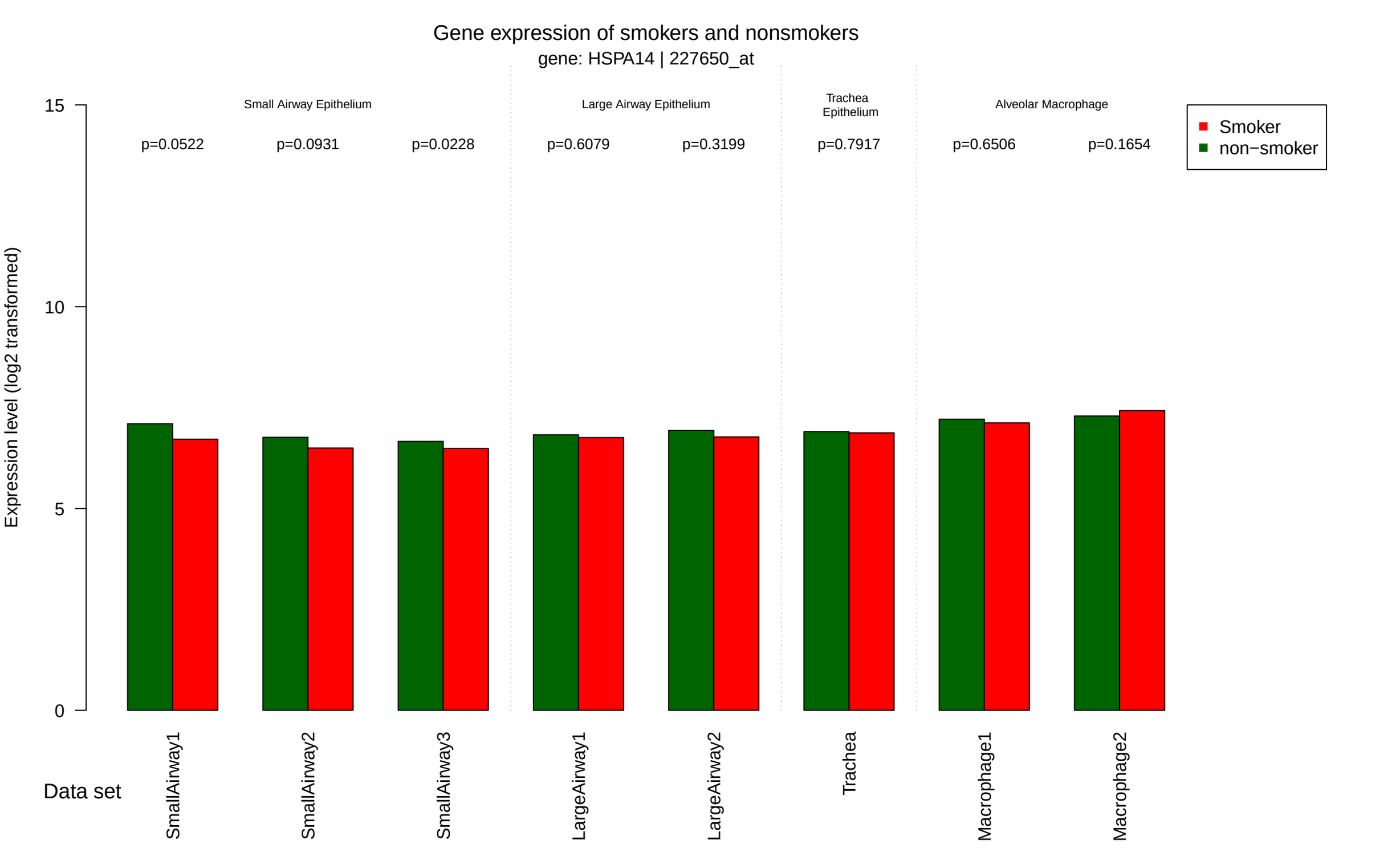
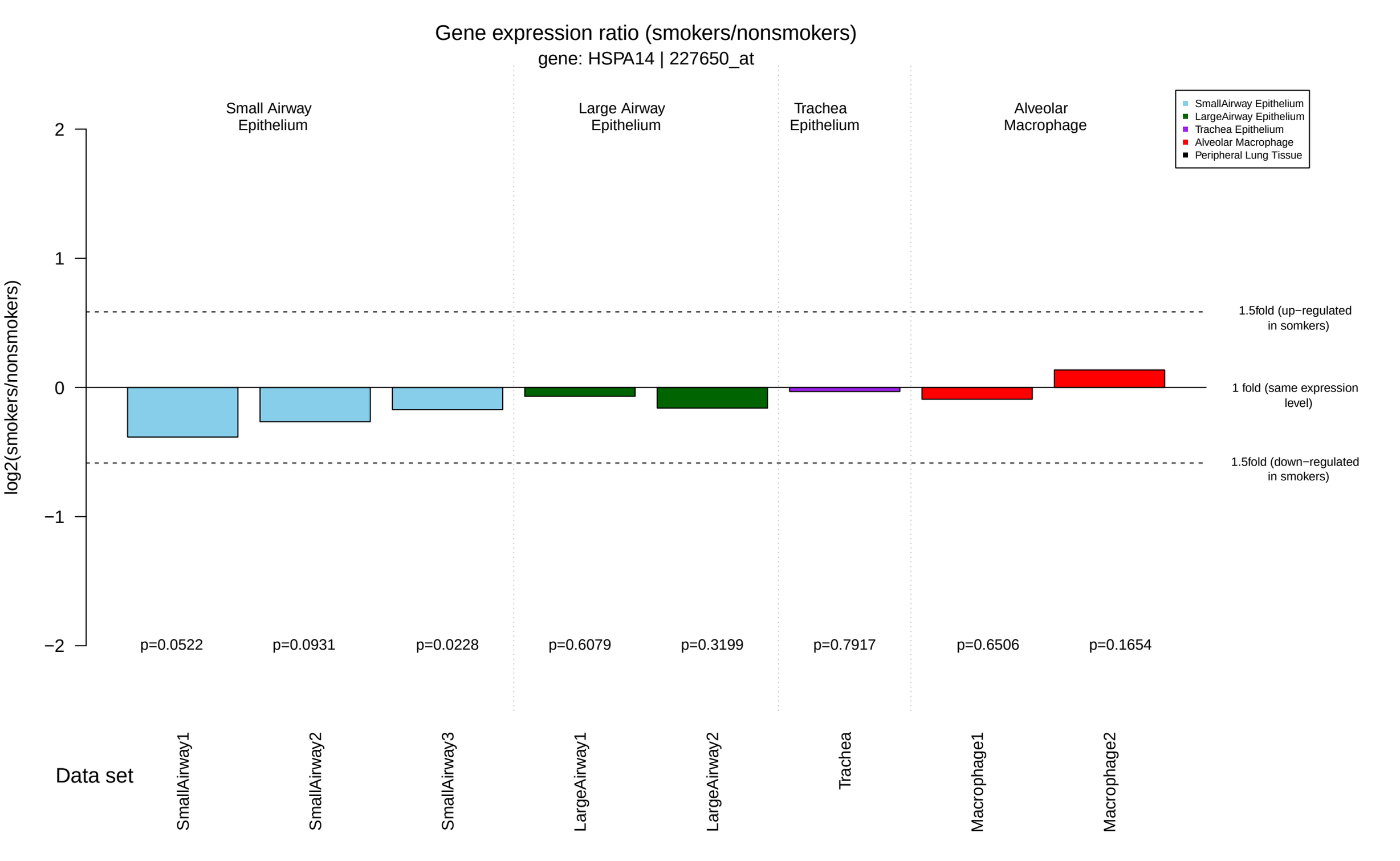
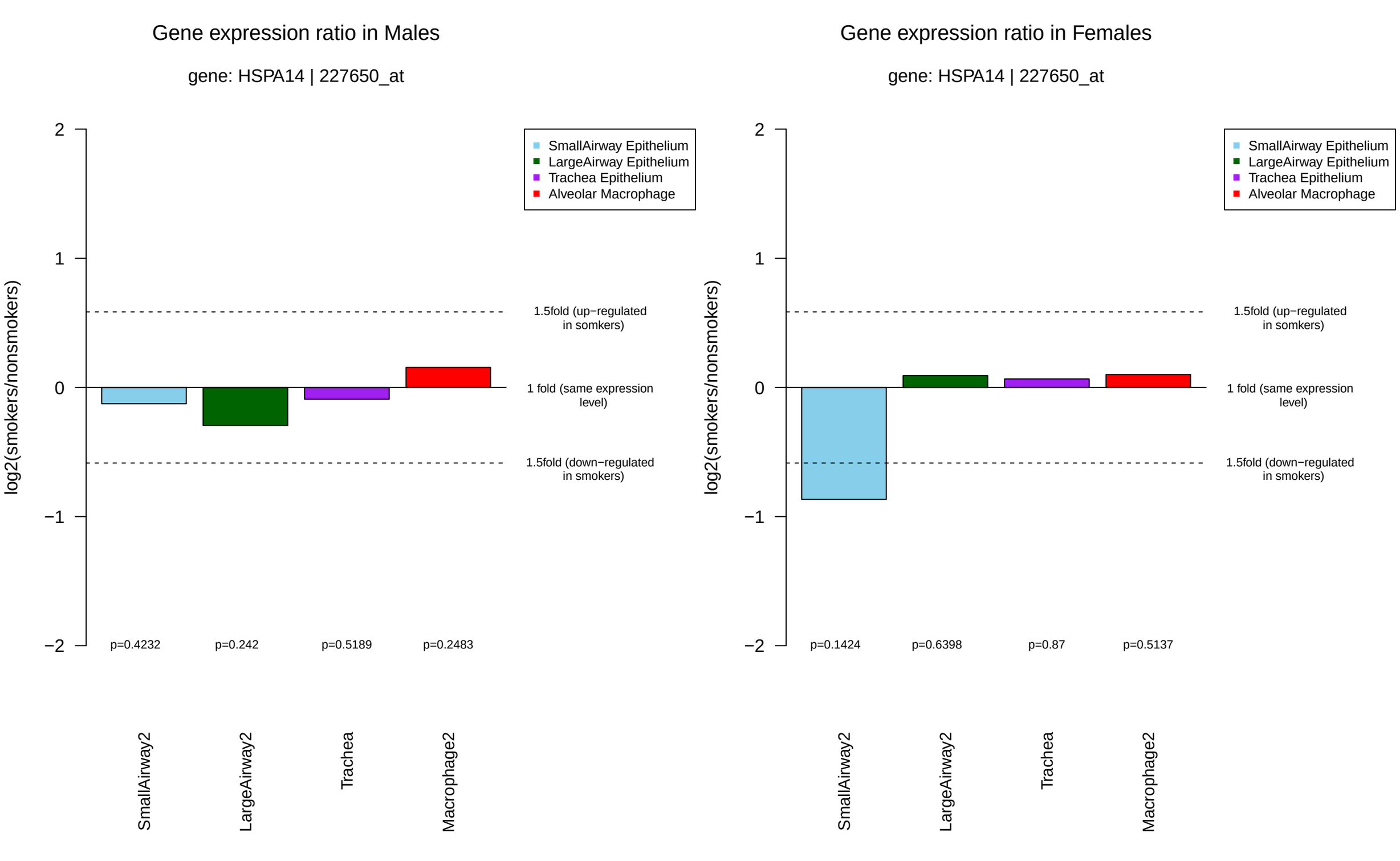
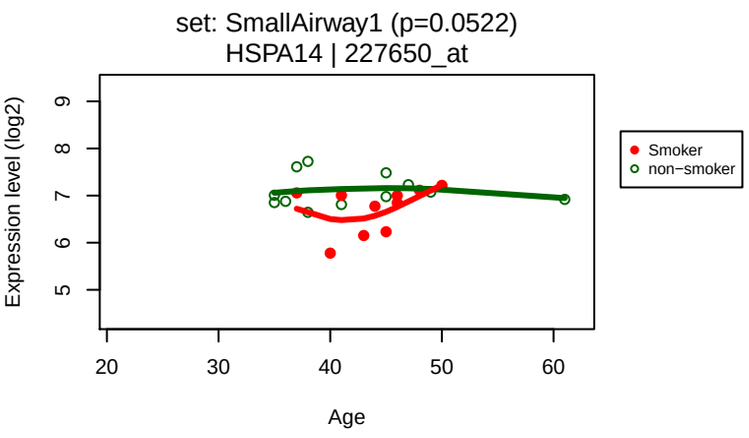
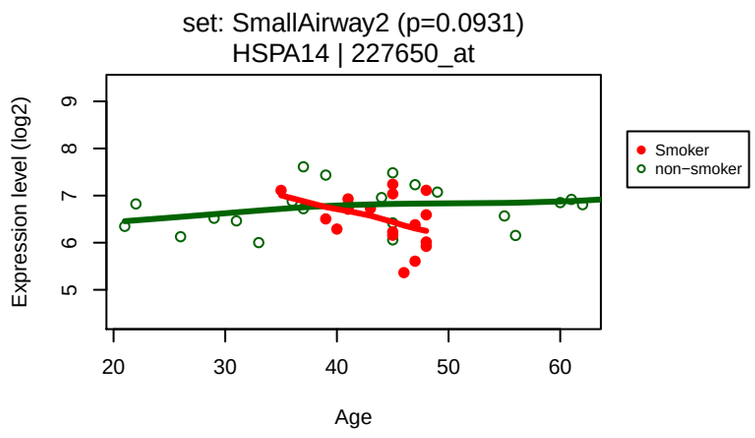
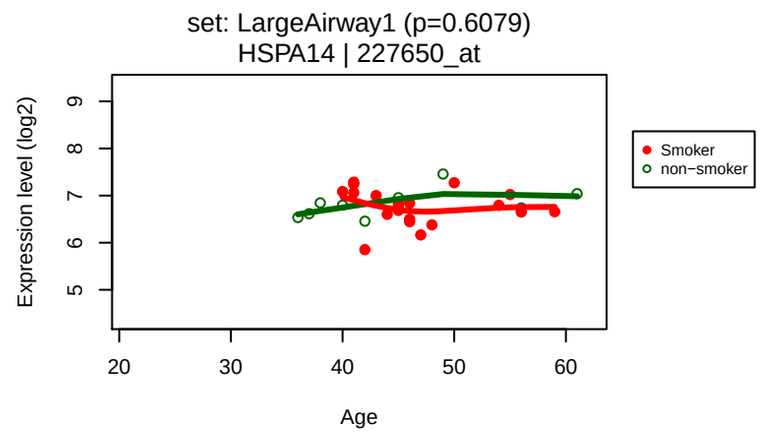

Gene expression of smokers and nonsmokers
gene: HSPA14 | 227650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0522
p=0.0931
p=0.0228
p=0.6079
p=0.3199
p=0.7917
p=0.6506
p=0.1654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

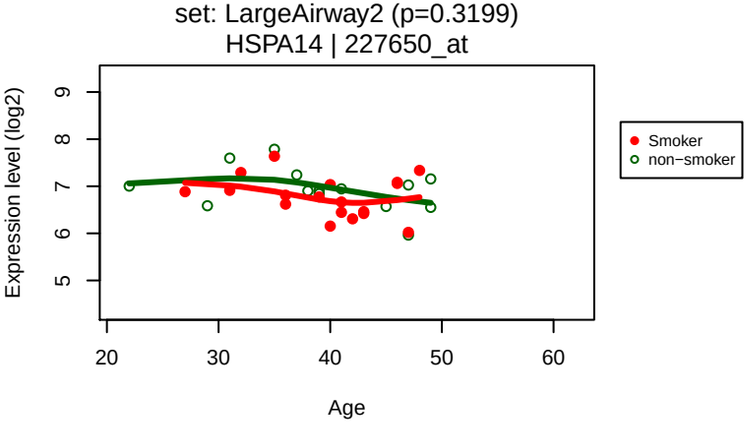

Gene expression ratio (smokers/nonsmokers)
gene: HSPA14 | 227650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0522
p=0.0931
p=0.0228
p=0.6079
p=0.3199
p=0.7917
p=0.6506
p=0.1654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

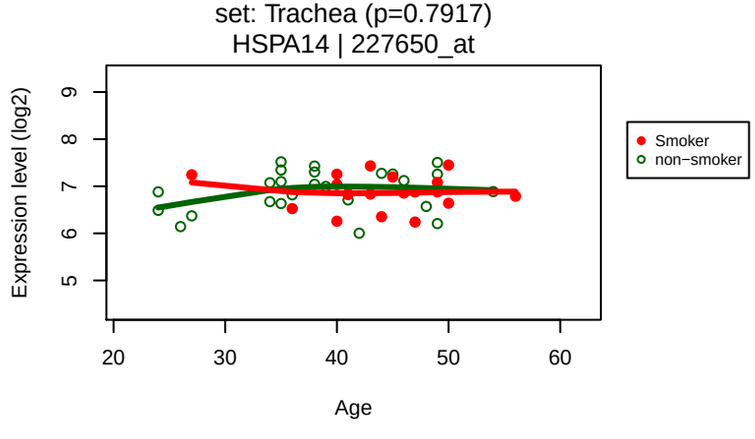

Gene expression ratio in Males
gene: HSPA14 | 227650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4232
p=0.242
p=0.5189
p=0.2483
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA14 | 227650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1424
p=0.6398
p=0.87
p=0.5137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

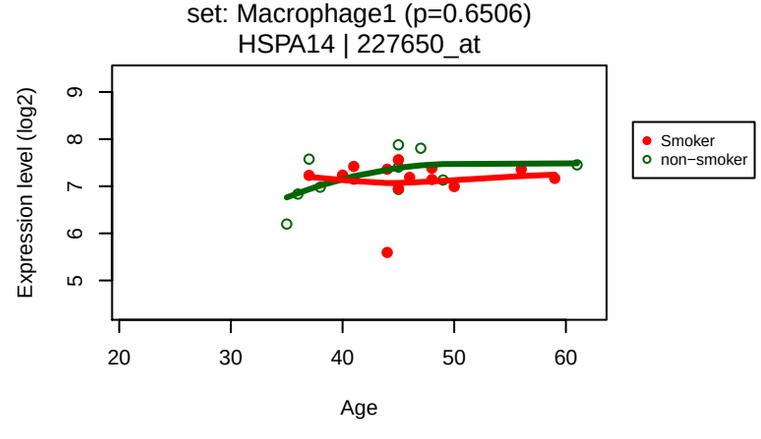

set: SmallAirway1 (p=0.0522)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker

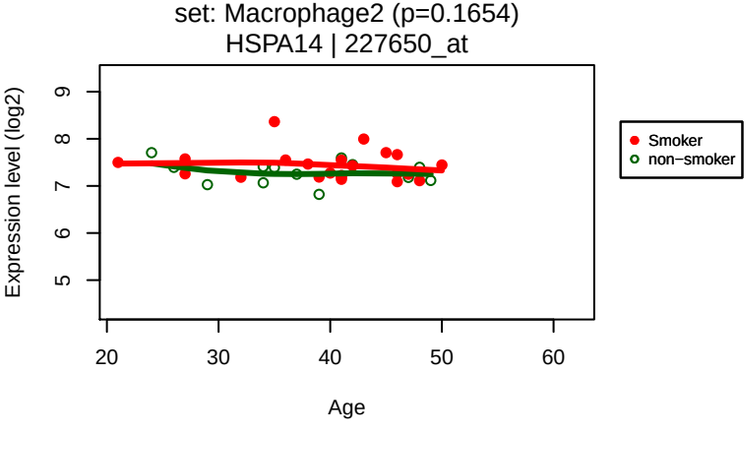

set: SmallAirway2 (p=0.0931)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker

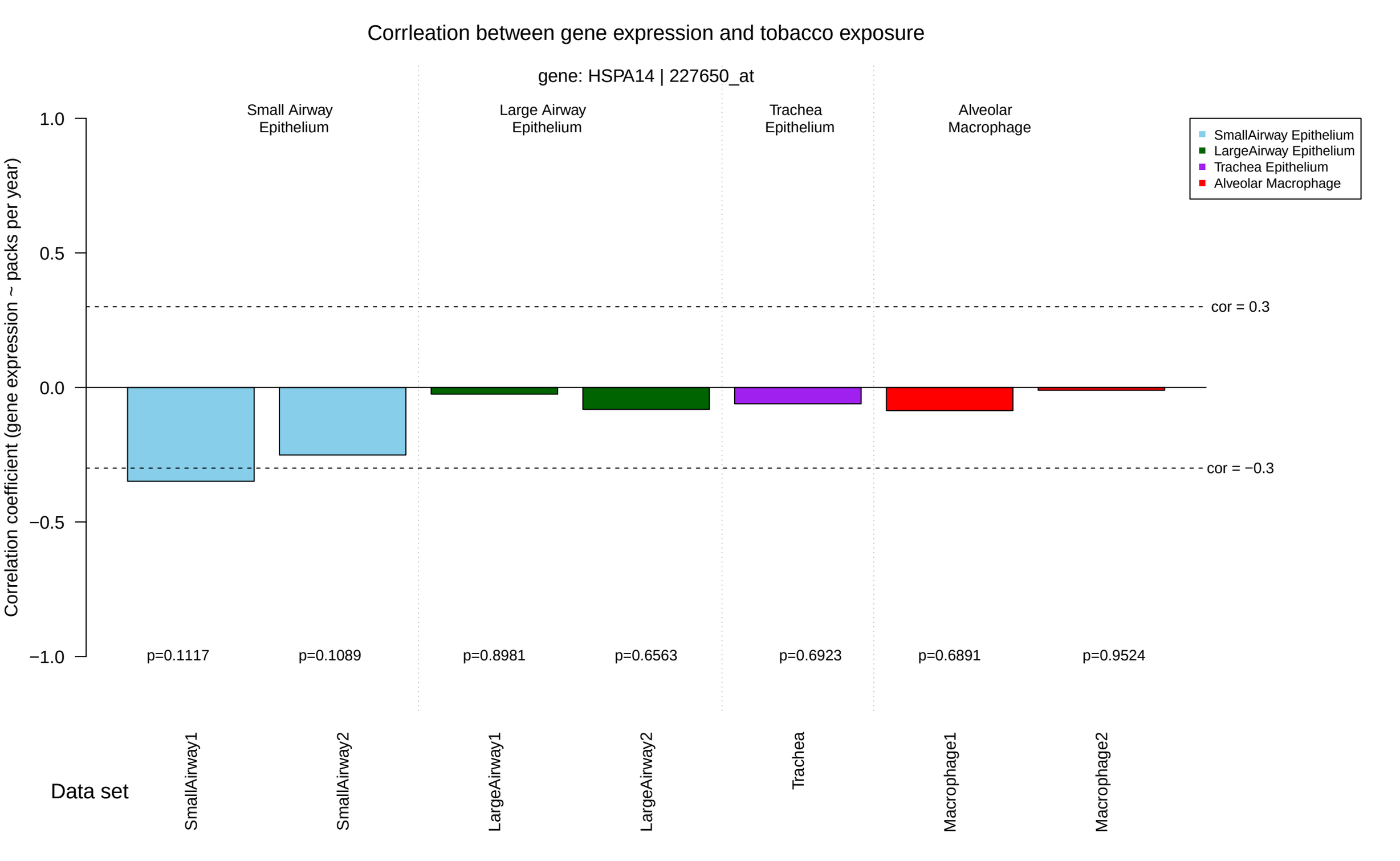

set: LargeAirway1 (p=0.6079)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3199)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7917)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6506)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1654)
HSPA14 | 227650_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA14 | 227650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1117
p=0.1089
p=0.8981
p=0.6563
p=0.6923
p=0.6891
p=0.9524
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2